



STAT 401: Mathematical Statistics

Probability Spaces, Asymptotic Theory, and Categorical Inference

Department of Statistics & Mathematical Sciences • Fall Semester 2026

Instructor: Prof. Alex Morgan
Email: alex.morgan@university.edu
Office Hours: Tuesday 14:00–16:00

Lecture Hours: Mon/Wed 10:00–11:30
Room: Math Sciences Hall 402
Lab Session: Friday 14:00–16:00 (Room 105)

Course Overview & Objectives

This course provides an advanced introduction to mathematical probability and statistical inference. Students develop rigorous mathematical foundations in measure-theoretic probability basics, characteristic functions, Central Limit asymptotics, maximum likelihood estimation, Gaussian hypothesis testing, and contingency table association.

Grading Scheme

Assessment Component	Weight	Description
Weekly Problem Sets (6)	30%	Mathematical derivations and Python simulations
Midterm Examination	30%	In-class theoretical and computational examination
Final Project / Exam	35%	Empirical dataset estimation and statistical inference report
Participation & Quizzes	5%	Active workshop and lab engagement

15-Week Curriculum Roadmap

- **Weeks 1–3:** Probability spaces, Kolmogorov axioms, conditional probability, Bayes’ rule.
- **Weeks 4–6:** Random variables, joint distributions, expectation, variance, covariance matrices.
- **Weeks 7–8:** Moment generating functions, Law of Large Numbers, Central Limit Theorem.
- **Week 9: Midterm Examination.**
- **Weeks 10–12:** Maximum likelihood estimation, asymptotic normality, Fisher information.
- **Weeks 13–15:** Categorical inference, contingency tables, χ^2 tests of independence, Odds Ratios.